

# **AI-compute supply chain: chart pack**

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, audited vs Atlas) and Atlas HS2012 annual.  
Sections: 1 the chain as flows (2024) -- 2 the time-varying factor model -- 3 concentration & fragmentation.  
Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-07-27

# 1. The chain as flows

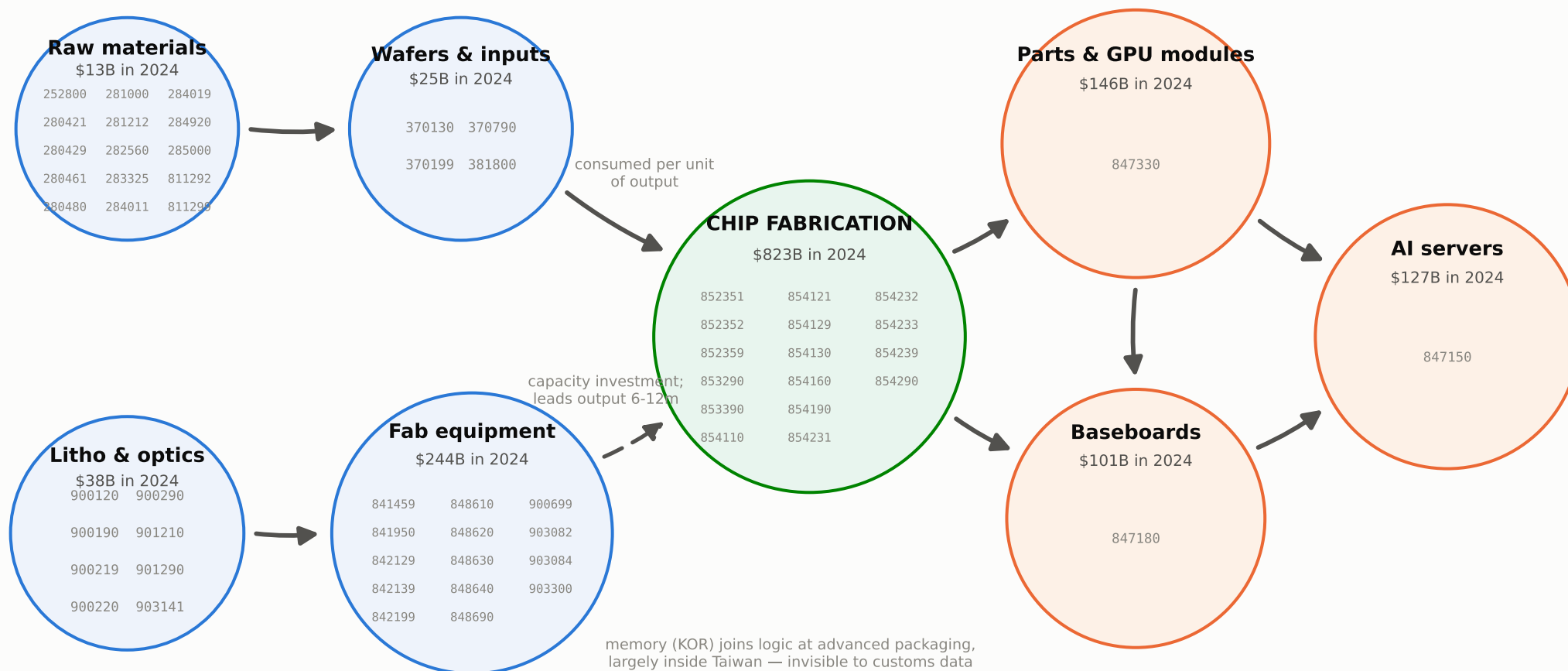
2024 bilateral flows, China+Hong Kong merged (CHK), intra-bloc excluded. Stages 1-5 from Atlas HS2012 annual data; stages 6-8 from the monthly panel (docs/data.md §3). Generated by scripts/export\_supply\_chain\_sankey.py.

# The chain's stylized shape, with HS6 codes per node

1. The chain as flows

Two parallel input branches (materials; tools) converge on chip fabrication; the output side is sequential. Node heights scale with the log of 2024 world trade. Solid arrows: consumed per unit of output; dashed: capacity investment (leads output 6-12 months in fab countries).

## The AI-compute supply chain — stylized shape, with the HS6 codes in each node



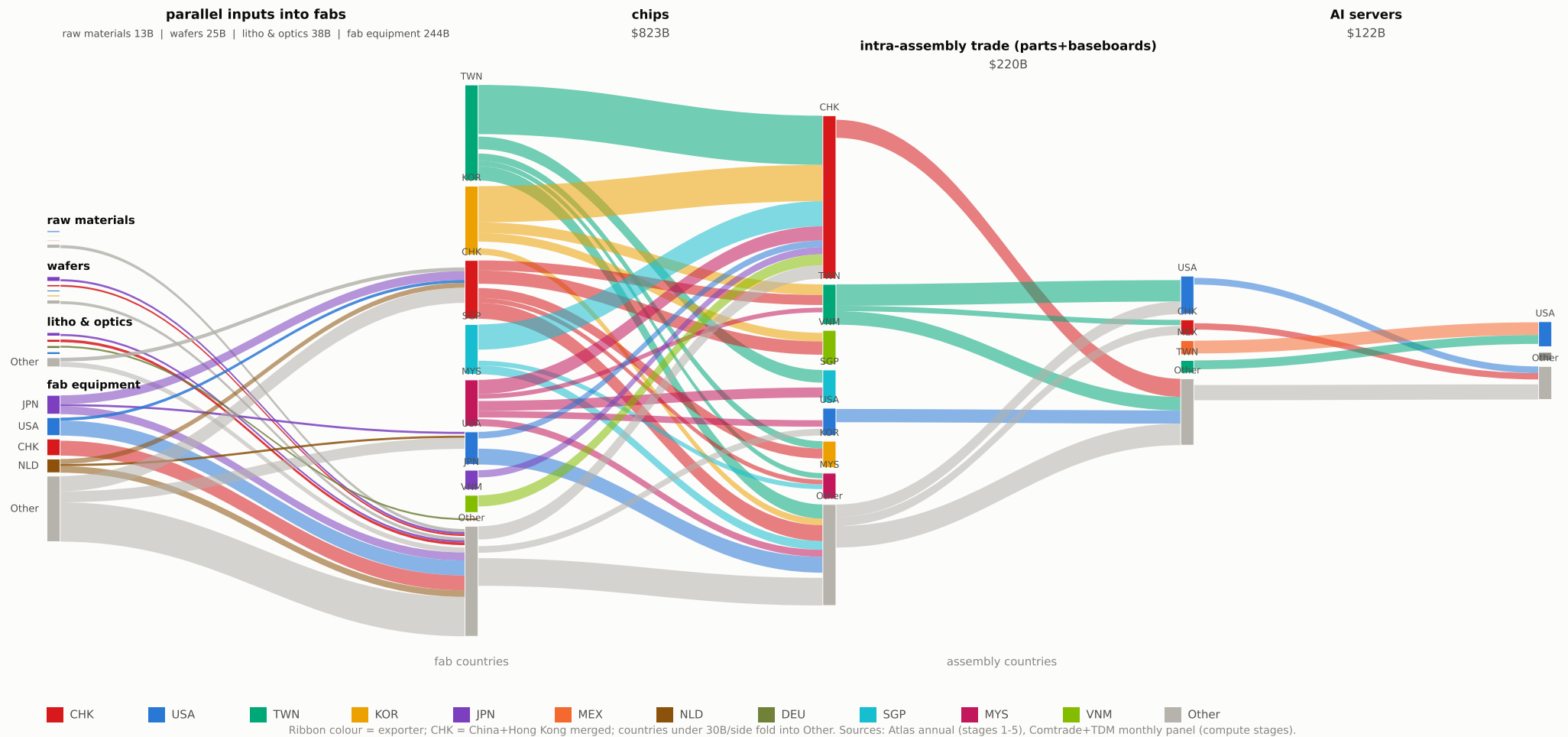
Two parallel input branches converge on fabrication; the output side is sequential. Circle areas follow the log of 2024 world trade (illustrative, so chips does not dwarf raw materials). Solid arrows: inputs consumed per unit of output. Dashed: capacity investment. Design/EDA/IP value enters as services, never as goods trade. Some small niche inputs sit outside the 60 codes (laser sources, vacuum pumps and valves, ABF substrates -- see notes/firm-level-supply-chain-data.md). Codes: OECD semiconductor mapping + Fed AI-compute basket (docs/data.md §1).

# Whole-chain overview, dollar scale (coarse)

1. The chain as flows

All eight stages left to right; band width = 2024 dollars. Inputs, litho optics and equipment run in parallel and converge on the fab countries; parts and baseboards form the intra-assembly segment before servers.

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)

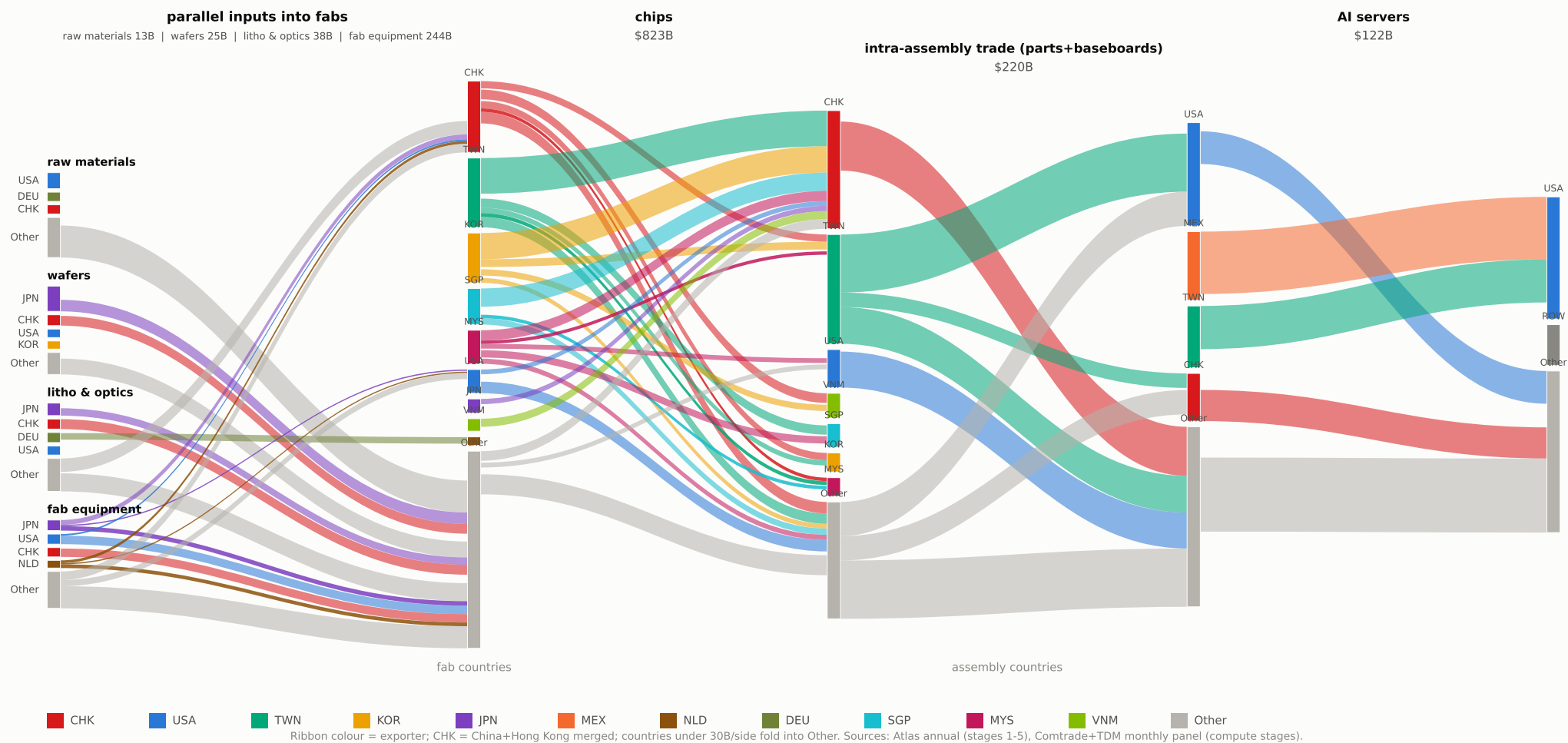


# Whole-chain overview, normalized (coarse)

1. The chain as flows

Same flows with each stage rescaled to equal width, so small stages' composition is readable (dollar scale makes chips dwarf everything).

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (stages equalized — shares only)

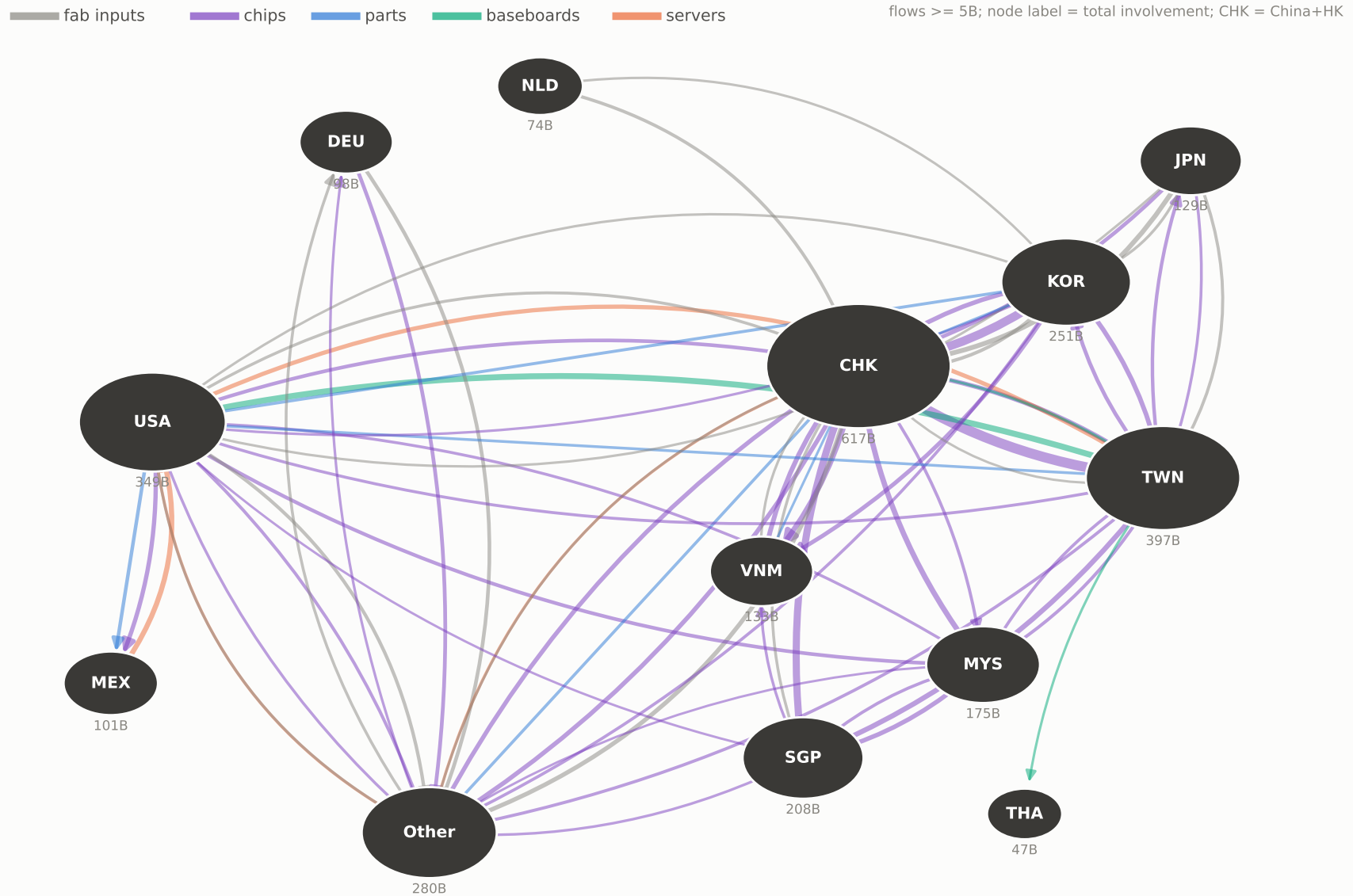


# Country network, all stages

1. The chain as flows

Nodes = countries, edges = 2024 flows colored by stage. Shows integrated roles a stage sequence cannot: Mexico takes in chips and parts and ships out servers at a single node.

The AI-compute supply chain as a country network, 2024 — edges by product stage, width ~ \$

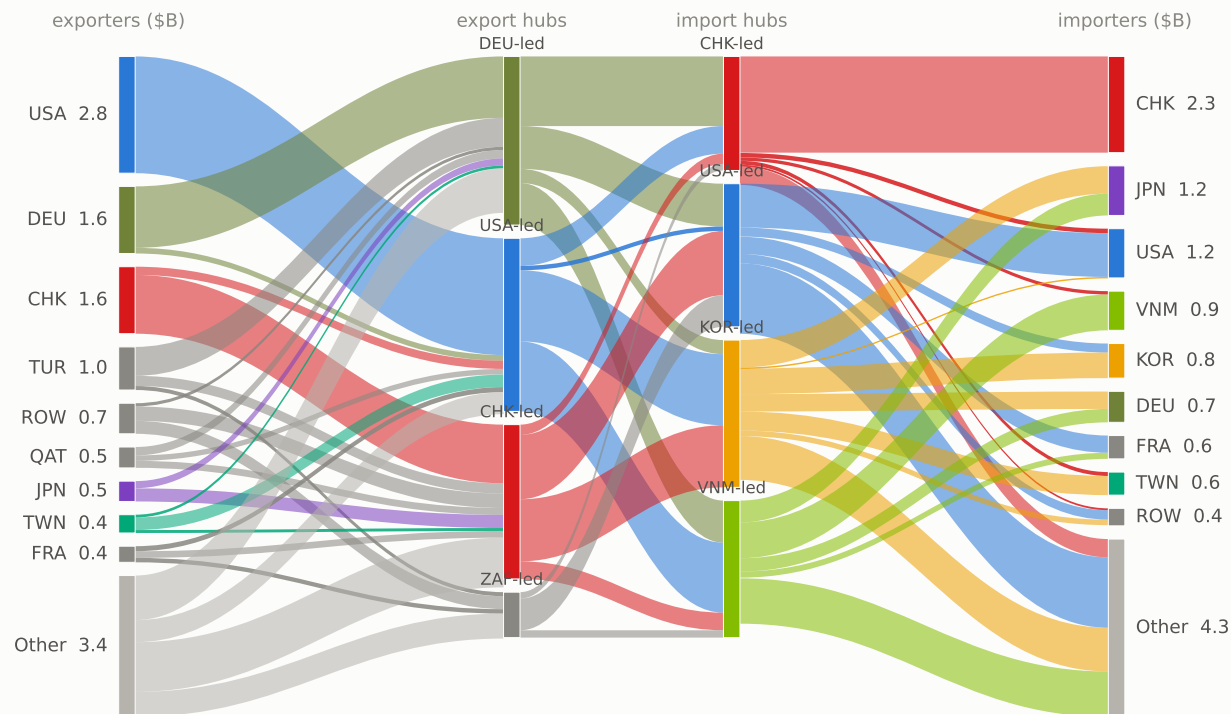


# Stage 1 -- raw materials, hub decomposition

1. The chain as flows

Four-column view: exporters -> export hubs -> import hubs -> importers, hubs estimated by MFM in the nonnegativity-identified basis. Applies to every stage page that follows; fits  $R^2$  0.95-0.99.

Stage 1 — Raw materials (silicon 280461, rare gases, Ga/Ge, chemicals) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$13B

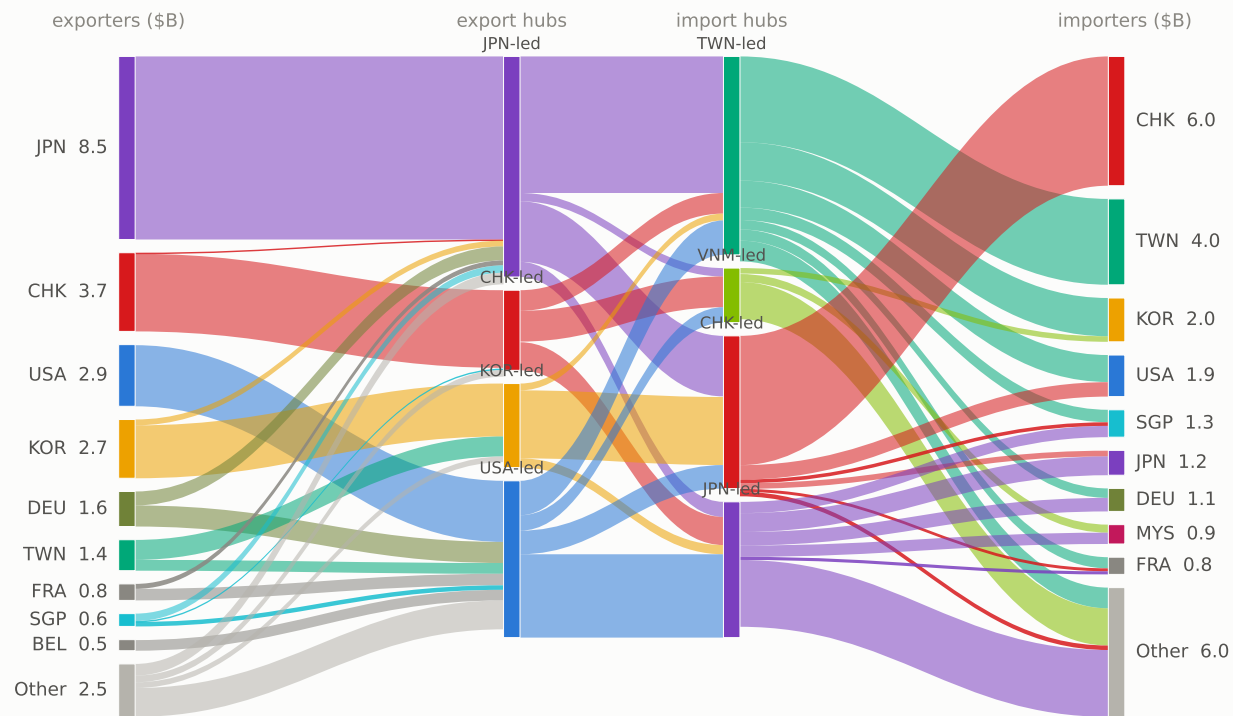


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

# Stage 2 -- wafers & wafer inputs

1. The chain as flows

Stage 2 — Wafers & wafer inputs (381800, 3701xx/370790) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$25B

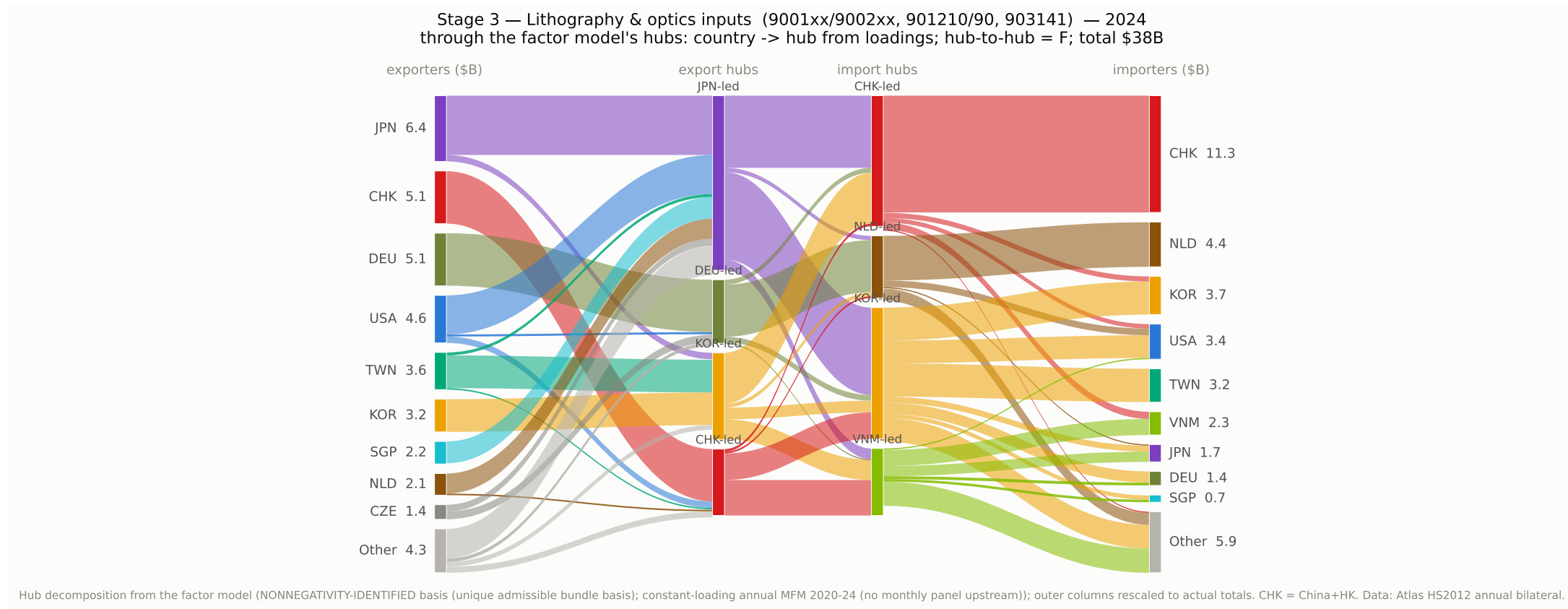


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.



# Stage 3 -- lithography & optics inputs

1. The chain as flows

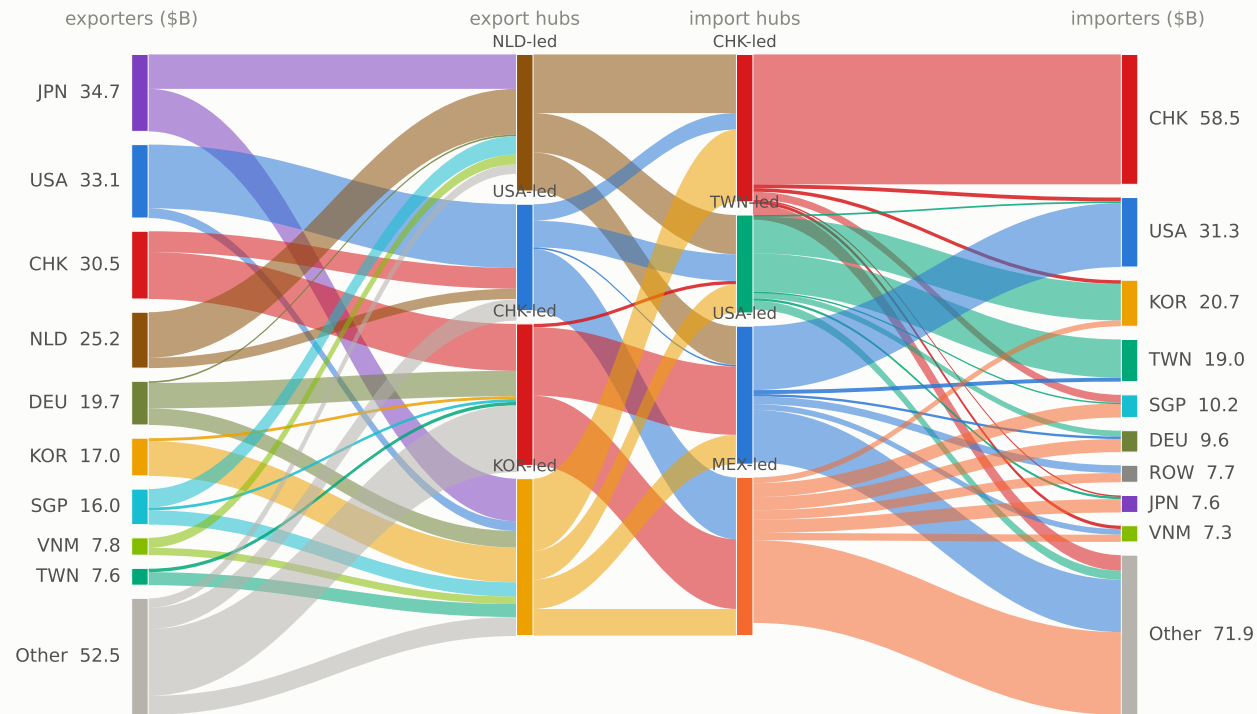


# Stage 4 -- fab equipment

1. The chain as flows

Capacity goods, not consumables: in fab countries these imports lead chip exports by 6-12 months (docs/notes/edge-type-tests.md).

Stage 4 — Fab equipment (8486xx, metrology, fab plant) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B

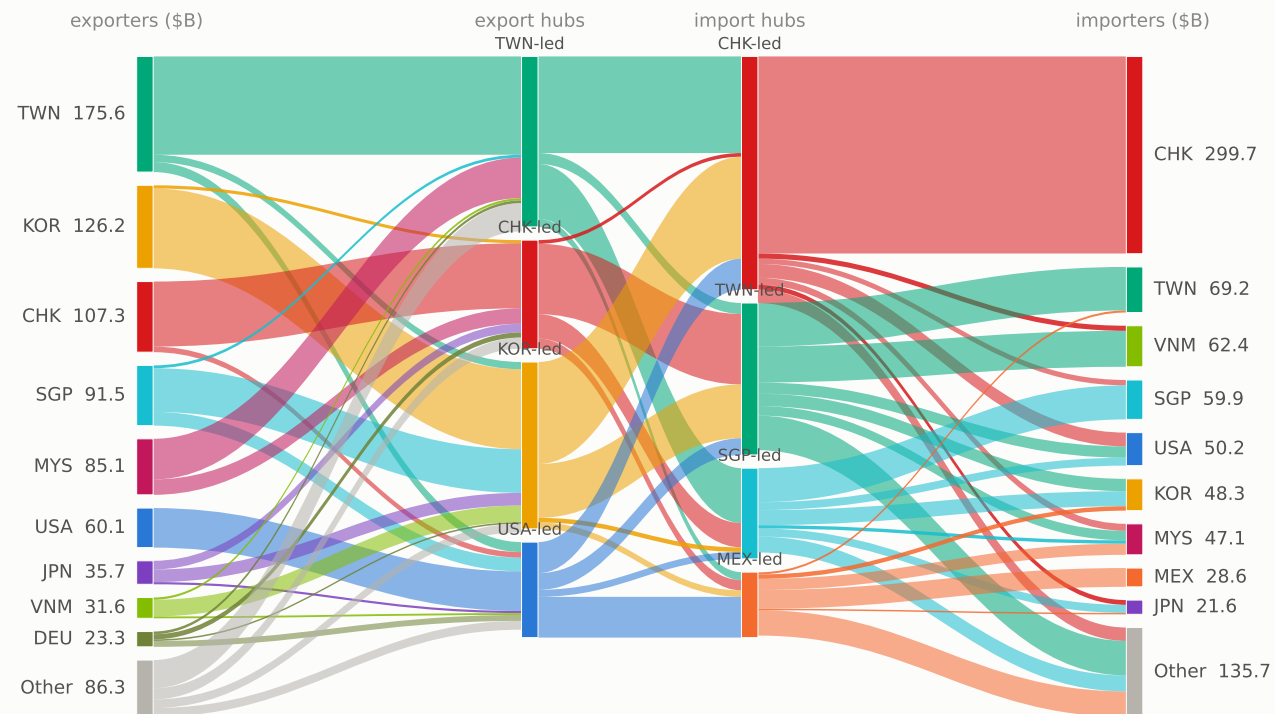


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

# Stage 5 -- chips (logic, memory, discretes)

1. The chain as flows

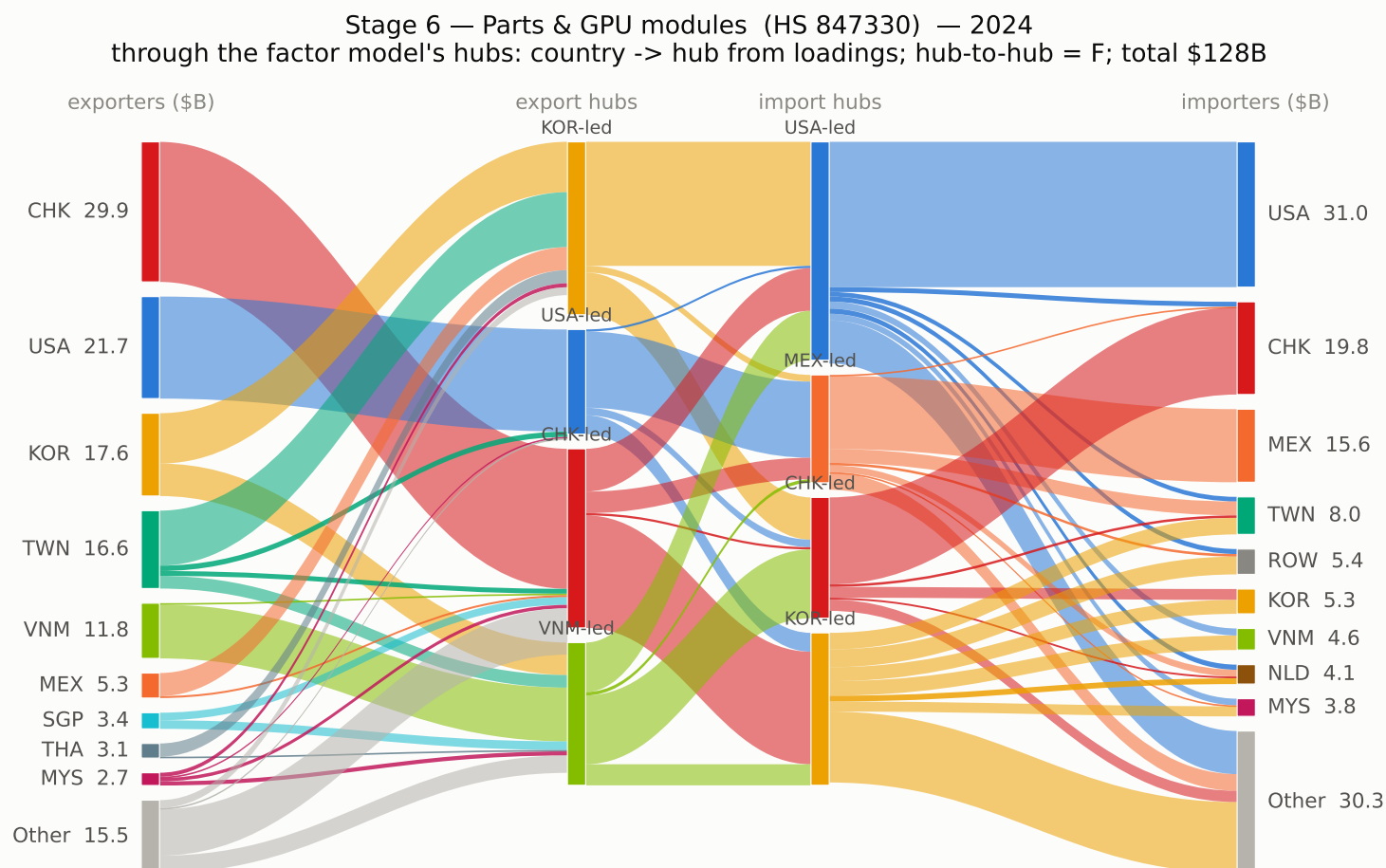
Stage 5 — Chips (logic, memory, discretes) (8542xx, 8541xx, media) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

# Stage 6 -- parts & GPU modules (847330)

1. The chain as flows

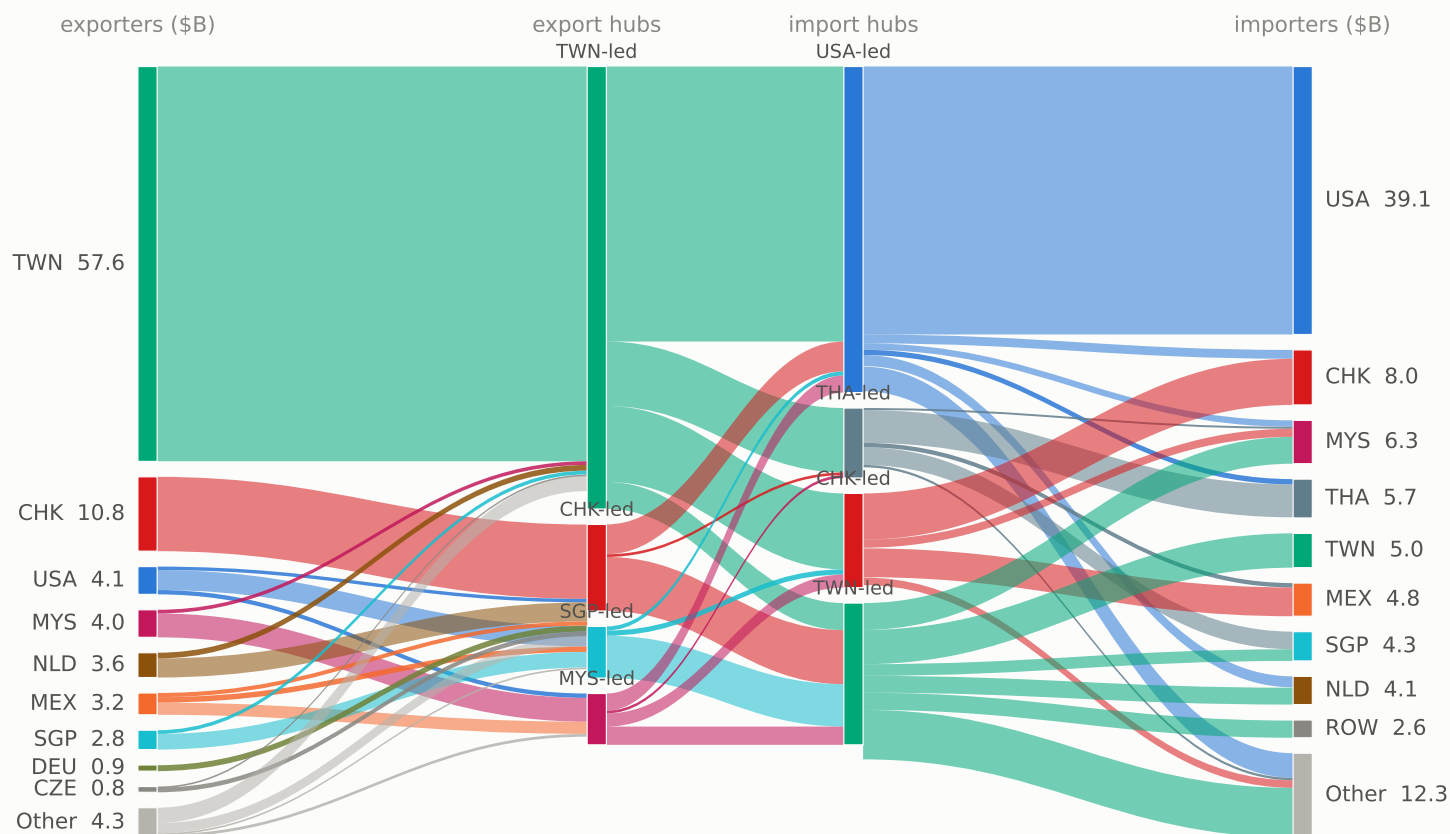


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

# Stage 7 -- baseboards (847180)

1. The chain as flows

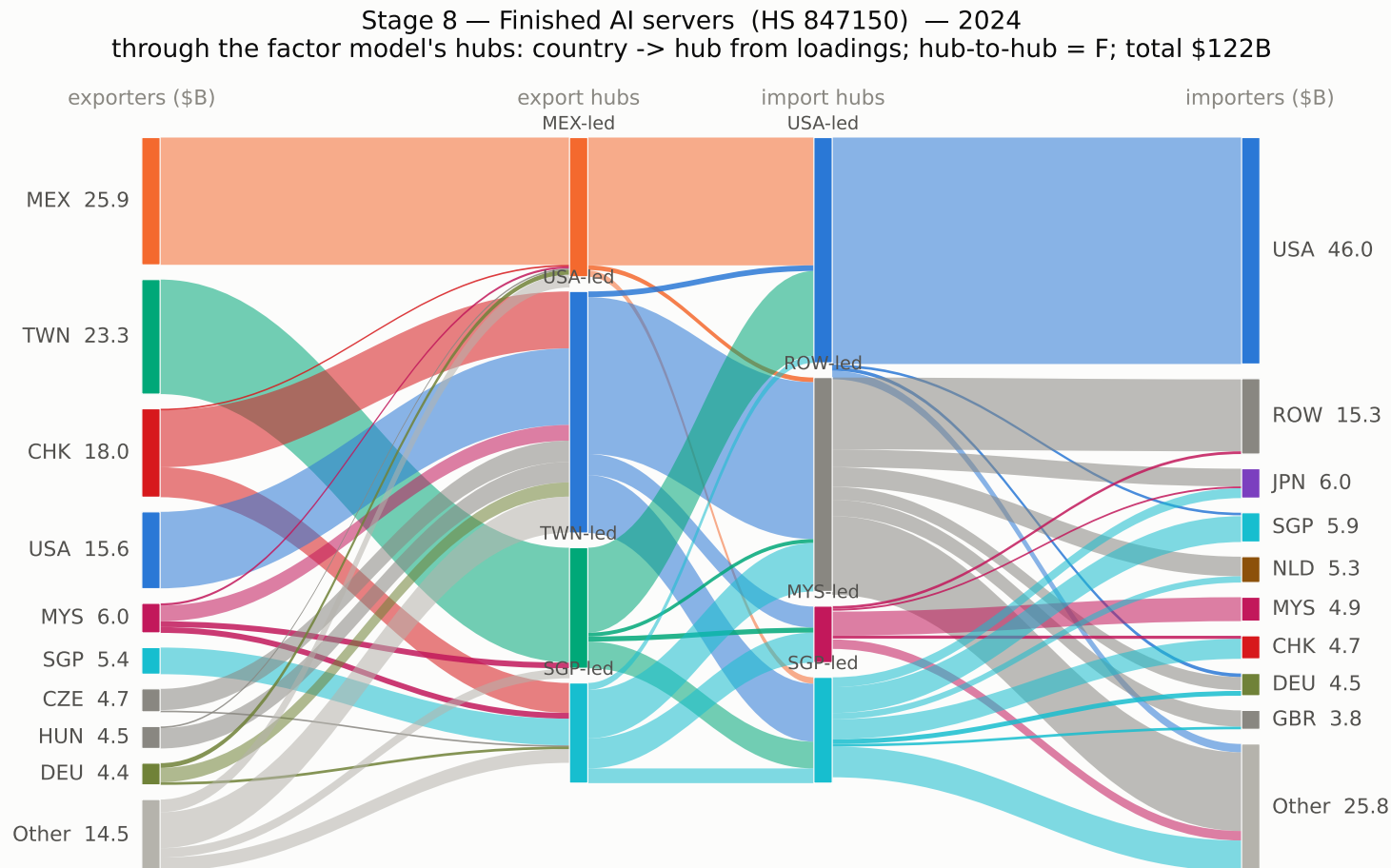
Stage 7 — Baseboards / other units (HS 847180) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$92B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

# Stage 8 -- AI servers (847150)

1. The chain as flows



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

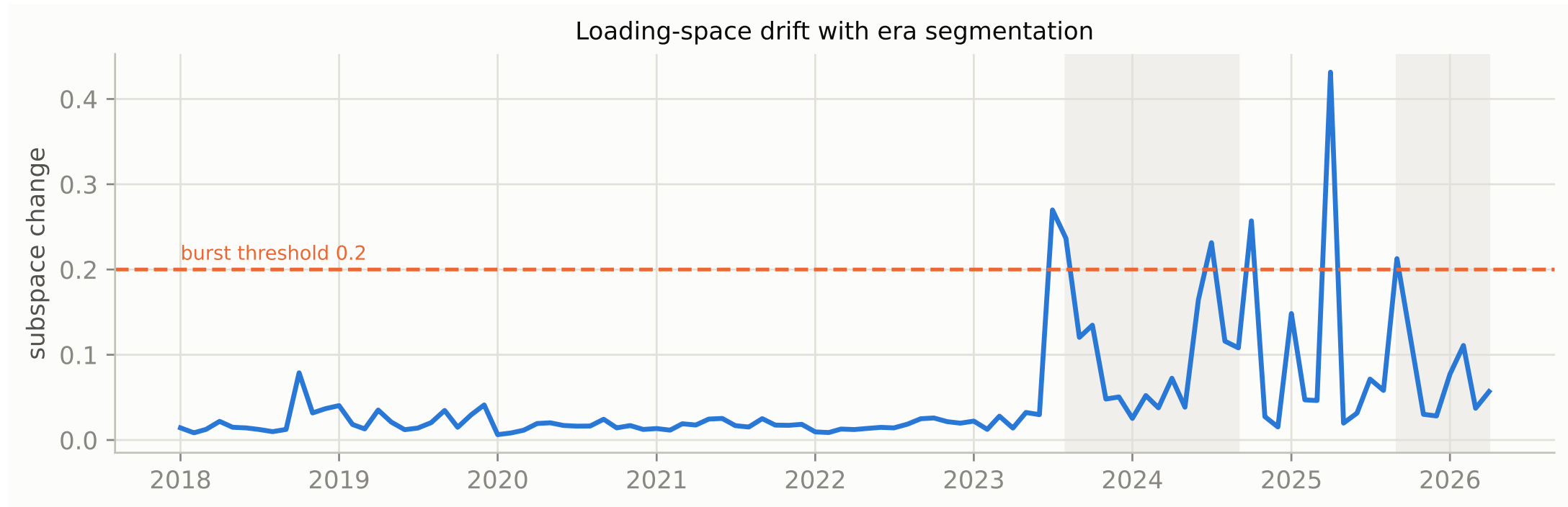
## **2. The time-varying factor model**

Era-anchored TV-MFM on the audited monthly panel (12m trailing window,  $k=4$ , \$B levels, NNF basis), 2017-12..2026-04. Shaded bands = eras between structural breaks. Generated by `scripts/tvmfm_monthly_anchored.py`; summaries in `results/mfm/tvmfm/`.

# AI compute, CHK basis -- loading drift and eras

2. The time-varying factor model

The preferred basis (entrepot churn netted; Taiwan separate). One unbroken 68-month era from 2017-12, then breaks at 2023-08, 2024-10, 2025-09.





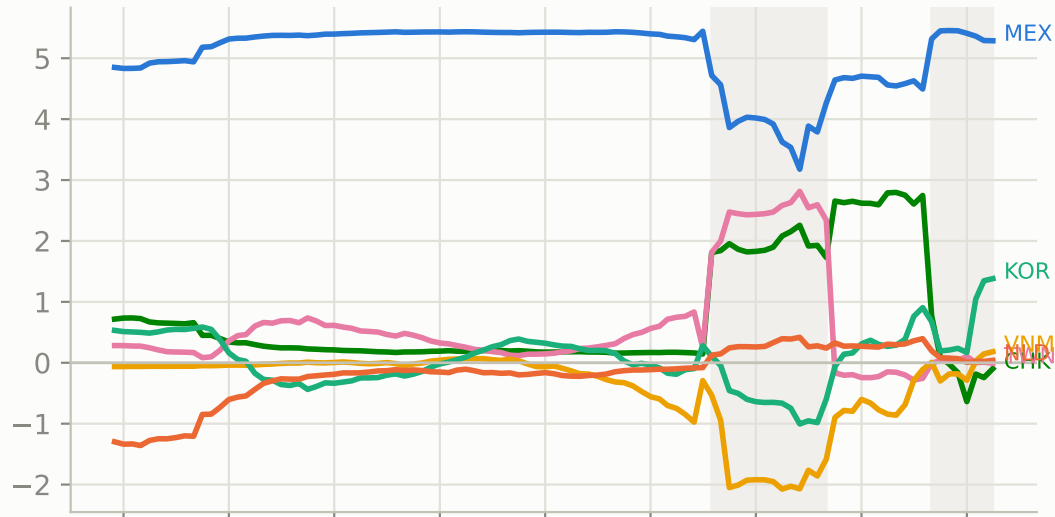
# AI compute, CHK basis -- export-hub loadings

2. The time-varying factor model

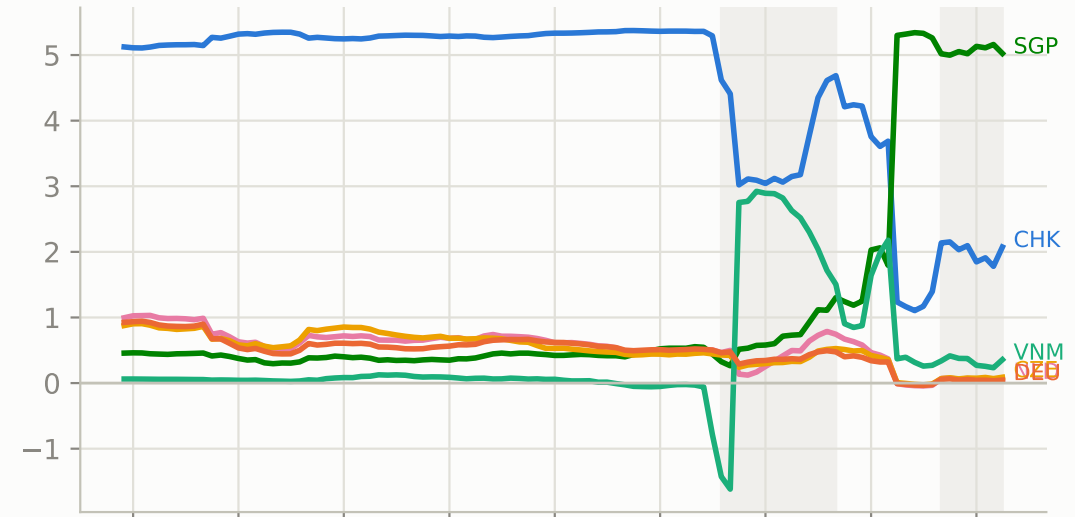
Per-hub country loadings through time. At 2024-10 the China-bloc hub loses its separate identity (cross-era similarity 0.45) and a Singapore-led hub appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

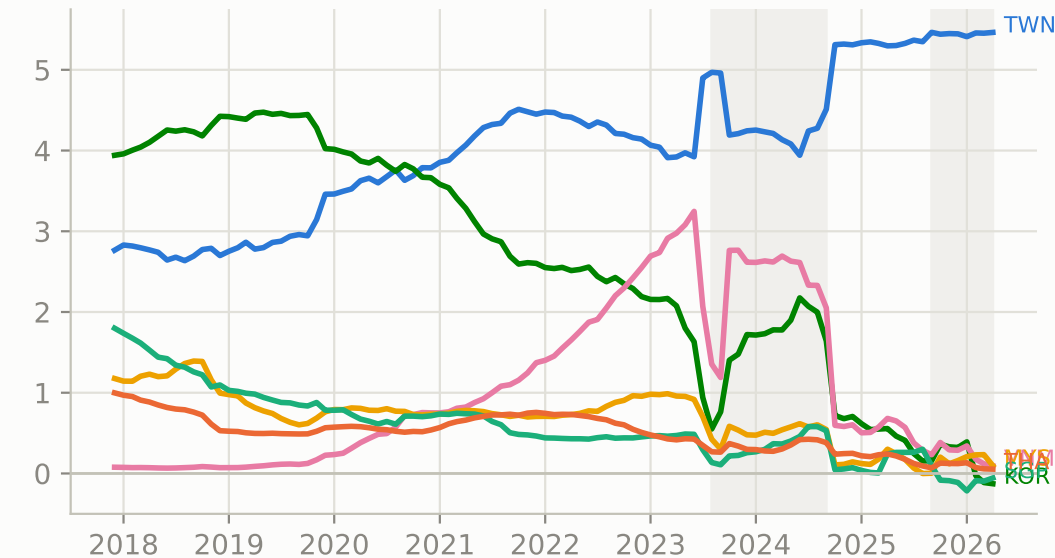
export hub 1



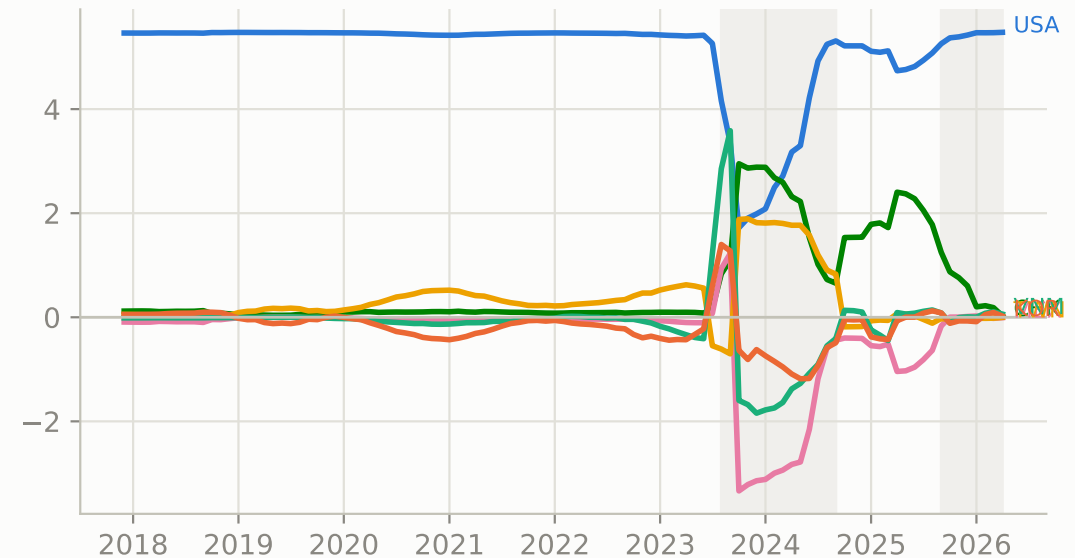
export hub 2



export hub 3



export hub 4



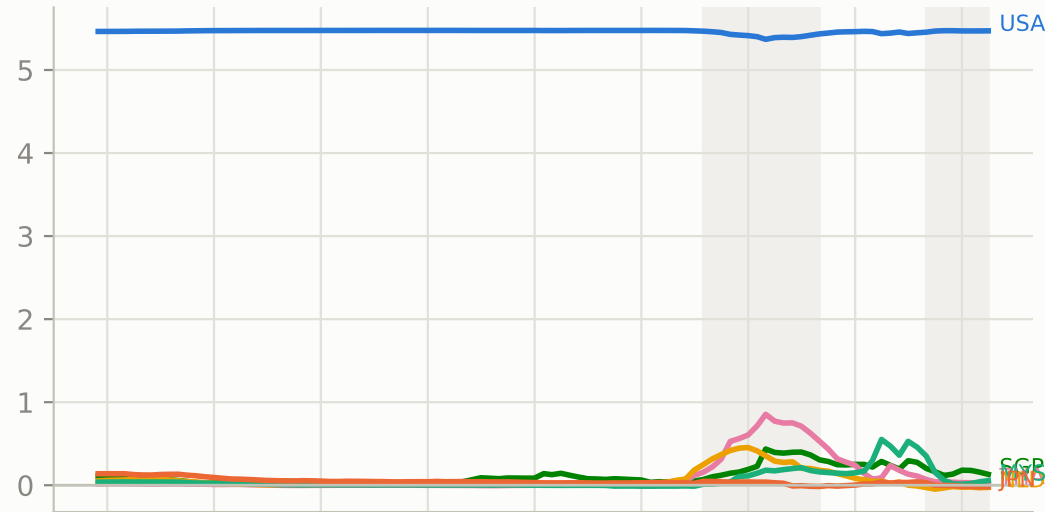
# AI compute, CHK basis -- import-hub loadings

2. The time-varying factor model

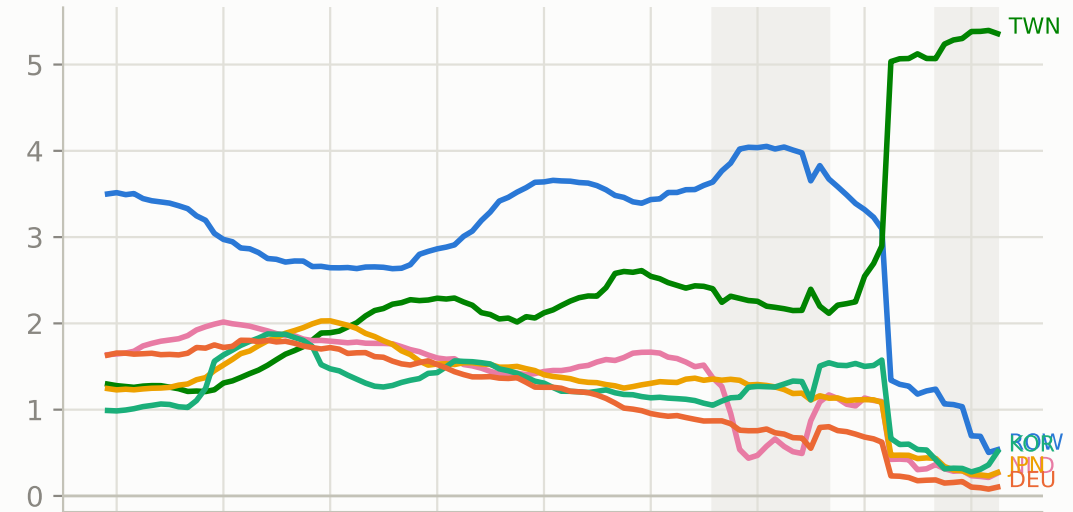
Import hubs are the stable side of the network (USA, Mexico, Taiwan, rest-of-world led).

AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)

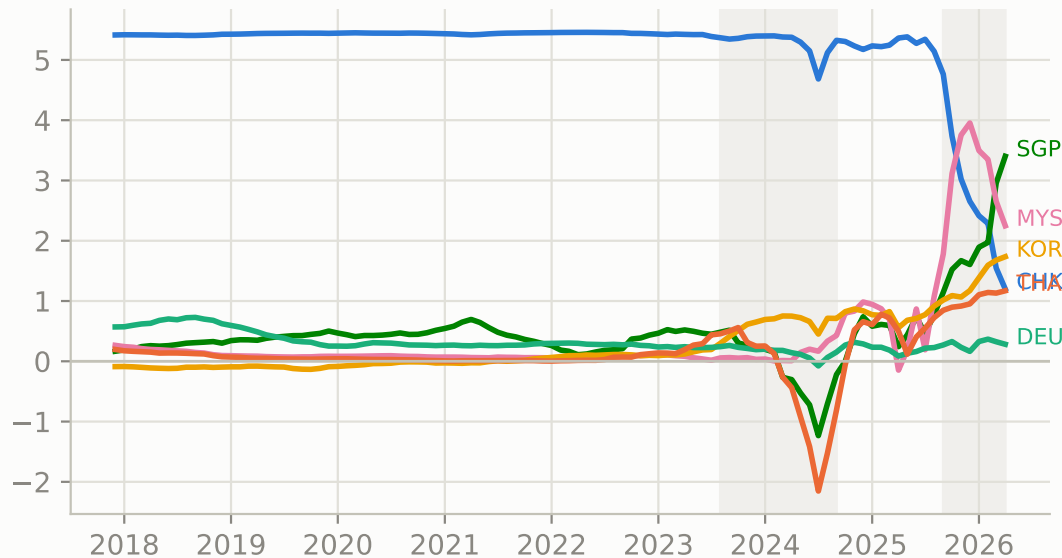
import hub 1



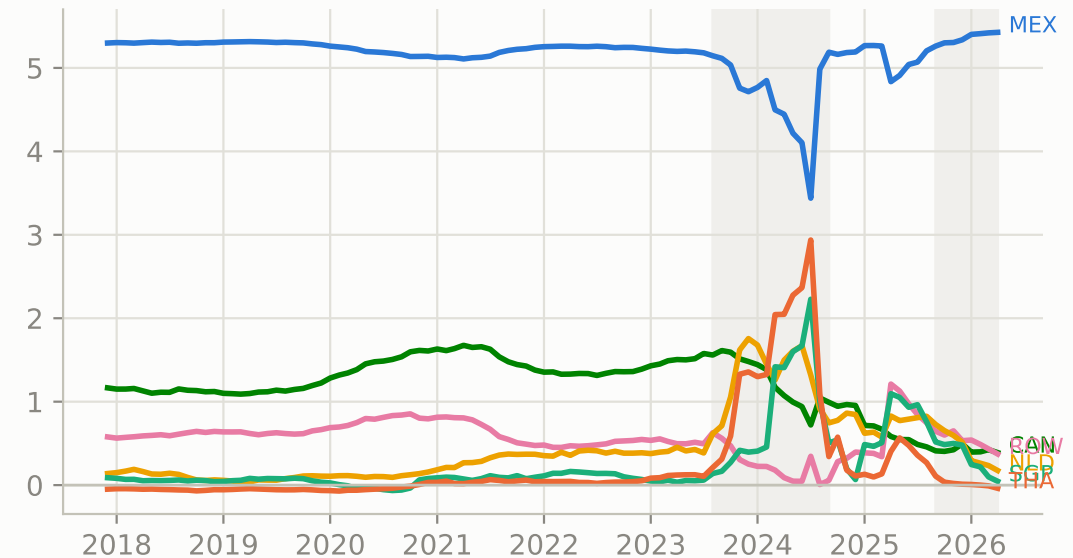
import hub 2



import hub 3



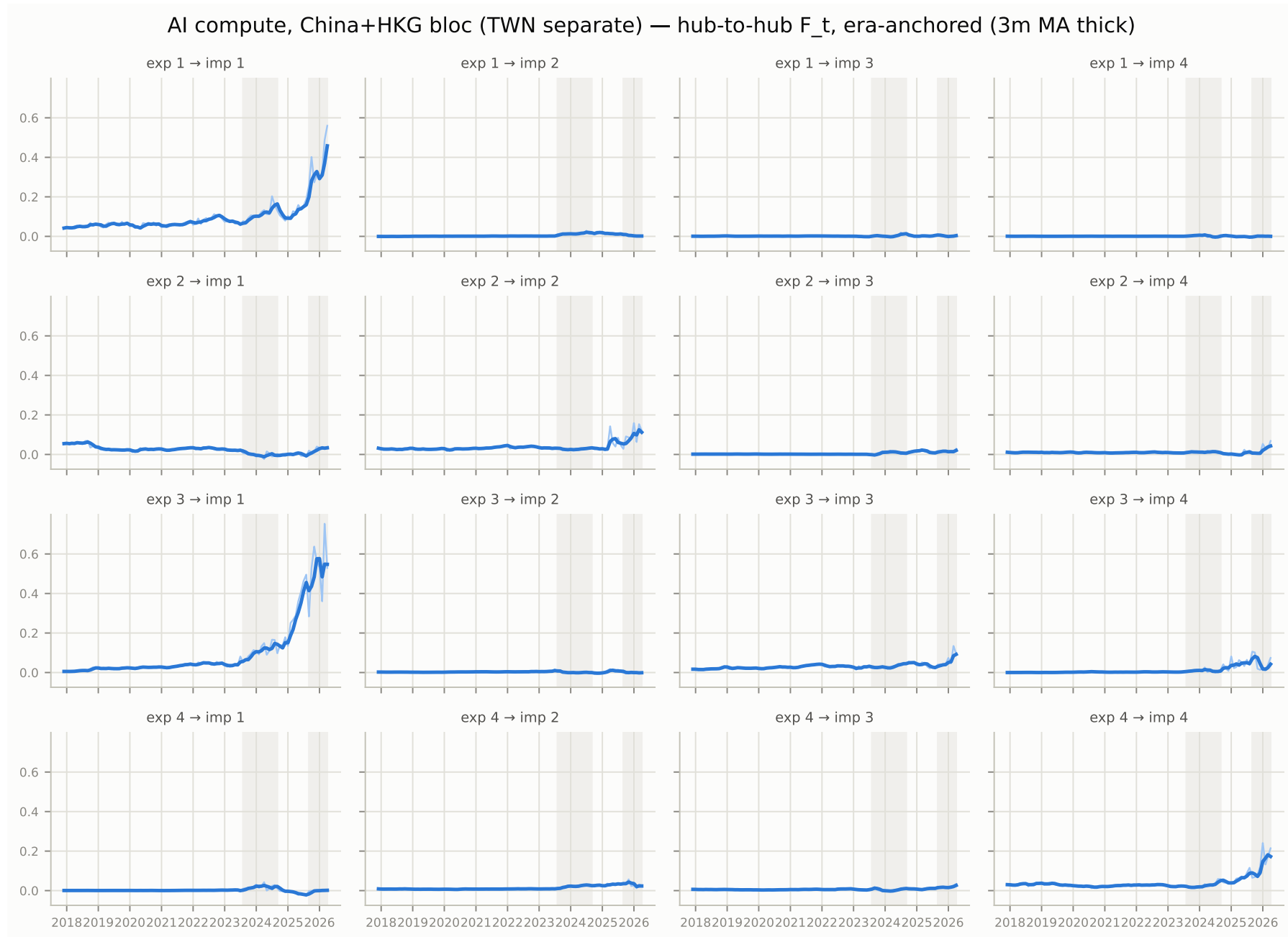
import hub 4



# AI compute, CHK basis -- hub-to-hub intensities $F_t$

2. The time-varying factor model

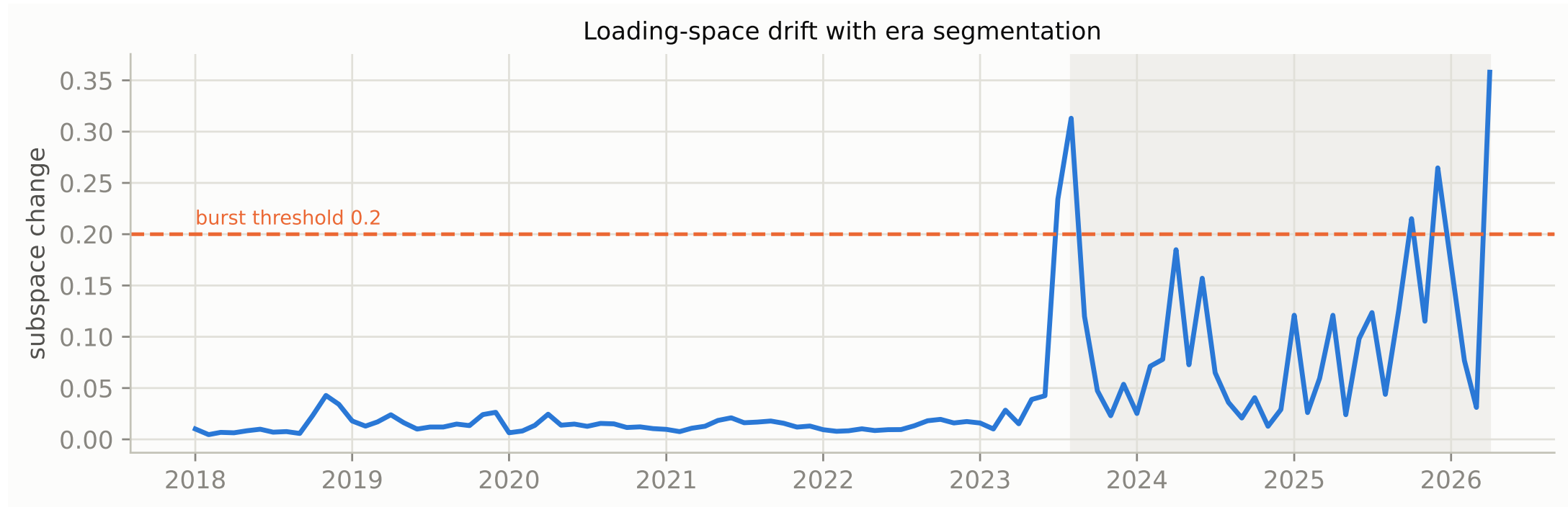
Each panel is one export-hub x import-hub channel in \$B (3m MA thick).



# AI compute, country level -- drift

2. The time-varying factor model

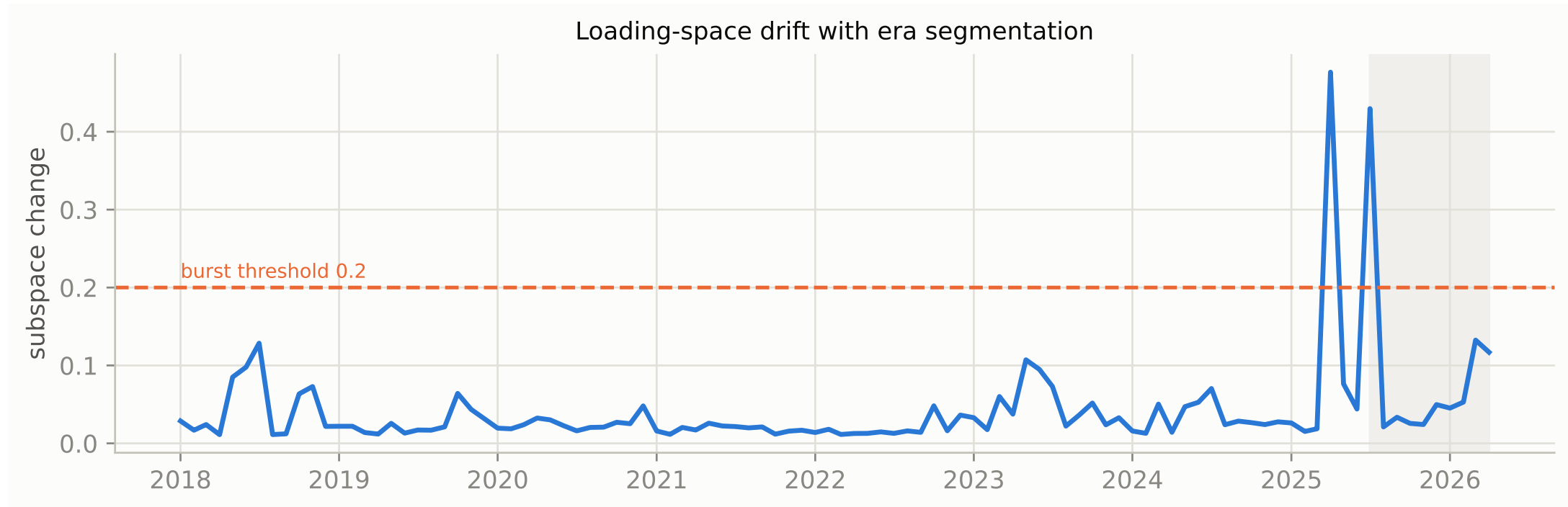
Without bloc merging the sample splits once: 2023-08. The demand event is the only dominant country-level break in nine years.



## Parts (847330), country level -- drift

2. The time-varying factor model

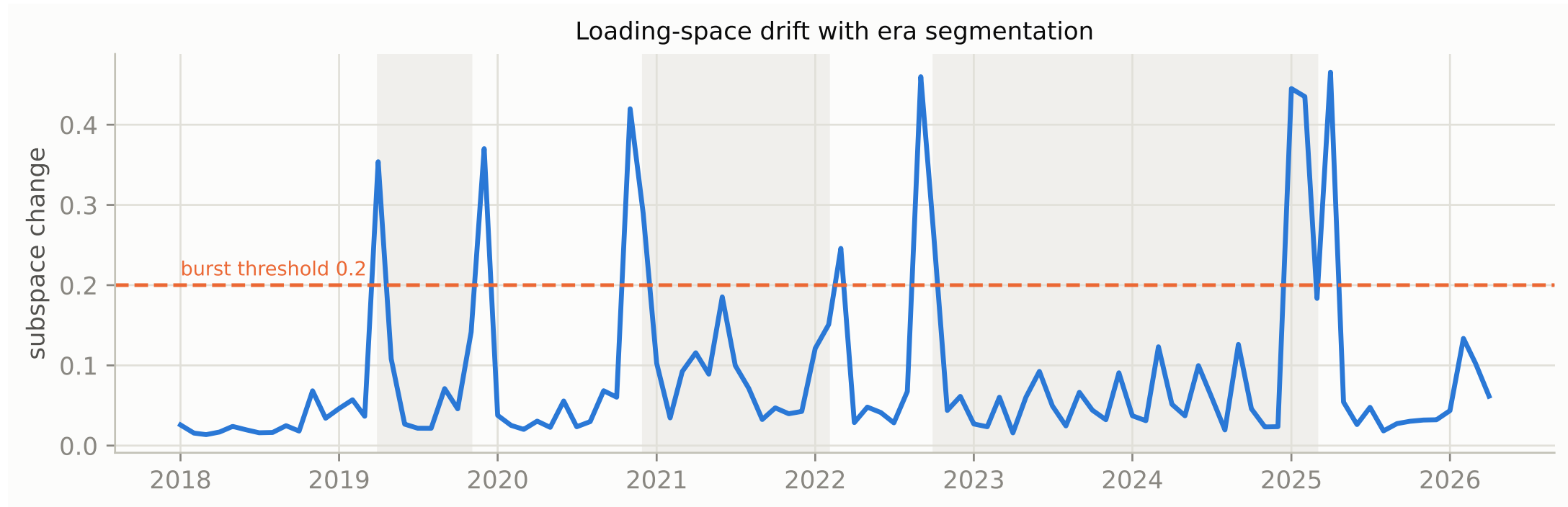
91 months without a structural break; the parts layer is the stable substrate of the network.



# AI compute, double-bloc basis -- drift

2. The time-varying factor model

CHK and USA+MEX both merged, isolating cross-bloc structure. Breaks at 2019-04/2019-12 (first tariff war), 2022-10 (first export controls), 2025-04 (tariffs): policy breaks blocs.



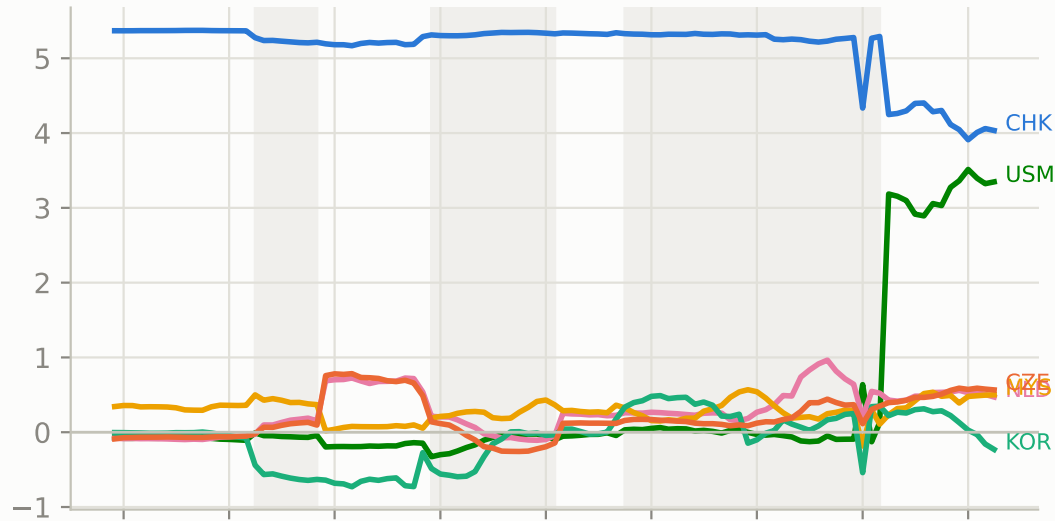
# AI compute, double-bloc basis -- export-hub loadings

2. The time-varying factor model

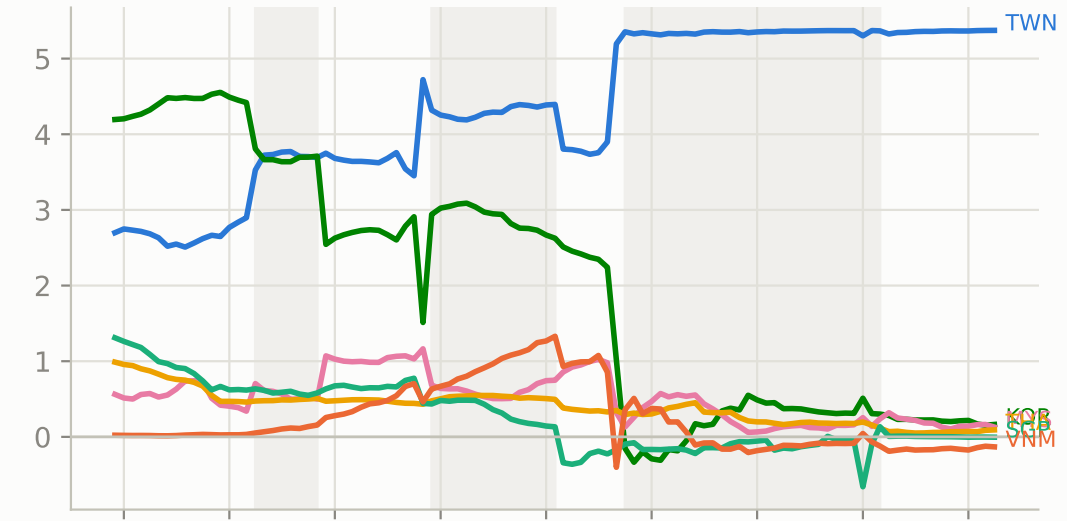
In the 2025-04 era the two blocs load together on one factor (CHK +3.97, USM +3.41) -- the entanglement signature unique to the tariff break.

AI compute, CHN+HKG and USA+MEX blocs — export loadings, era-anchored (12m window; shaded bands = eras)

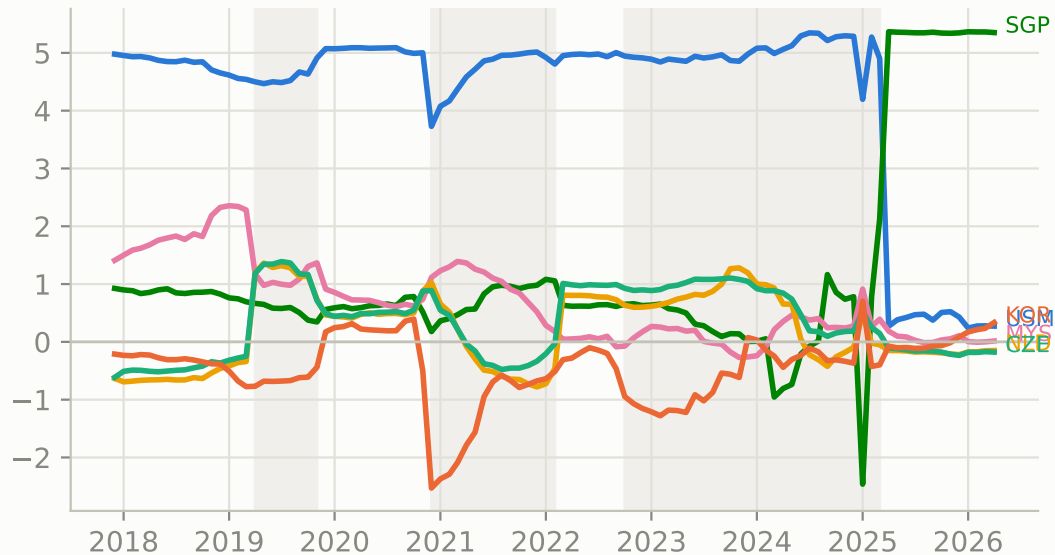
export hub 1



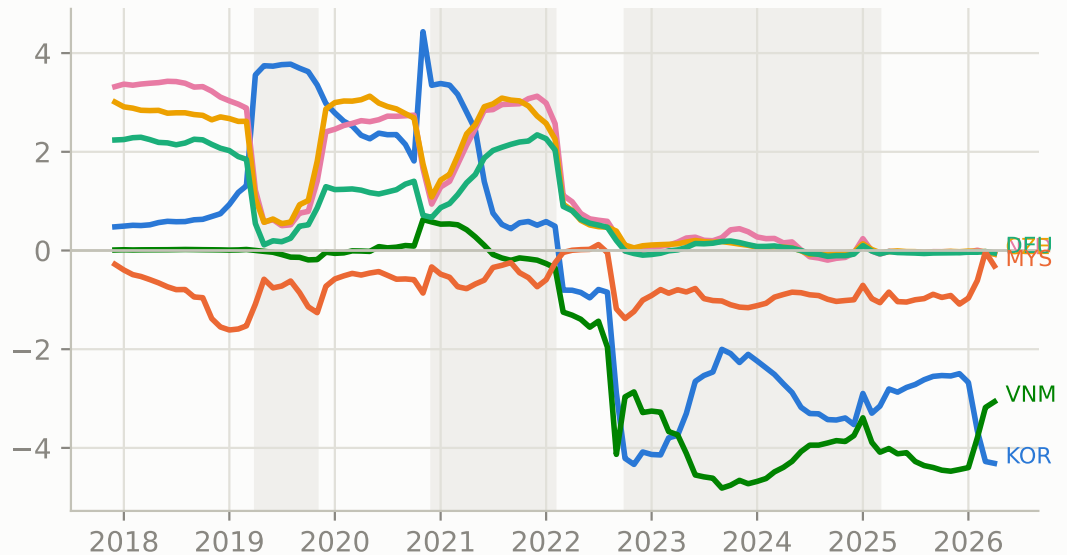
export hub 2



export hub 3



export hub 4



### **3. Concentration & fragmentation**

Measure system of docs/concentration-fragmentation-nnf-mfm-trade.md computed monthly from the TV-MFM (CHK basis). Dotted lines mark 2023-07 and 2025-04. Interpretation: results/network\_stats/findings.md. Generated by scripts/network\_stats.py.

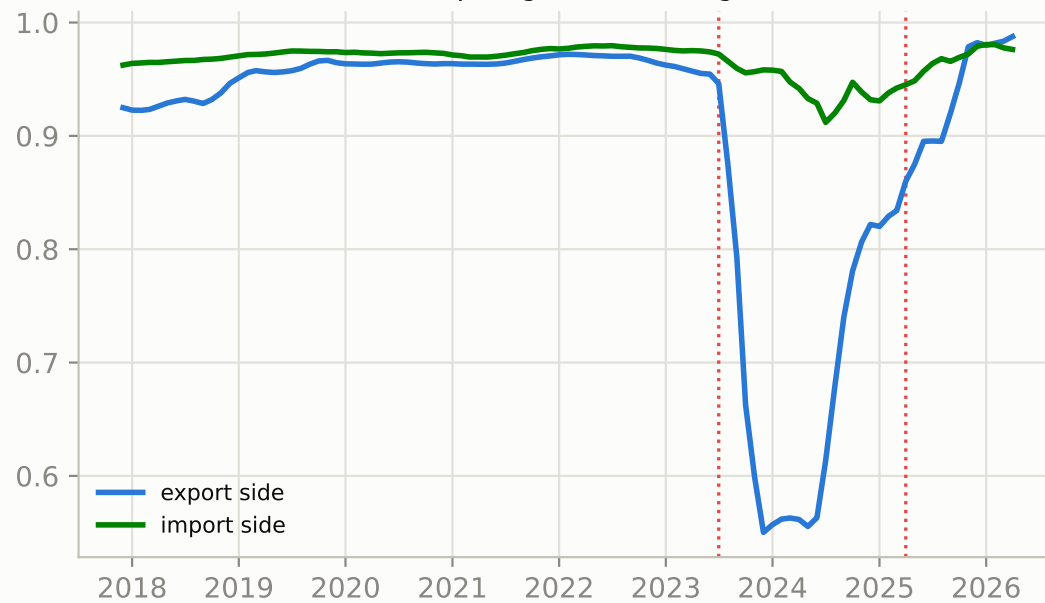


# Two notions of fragmentation -- AI compute

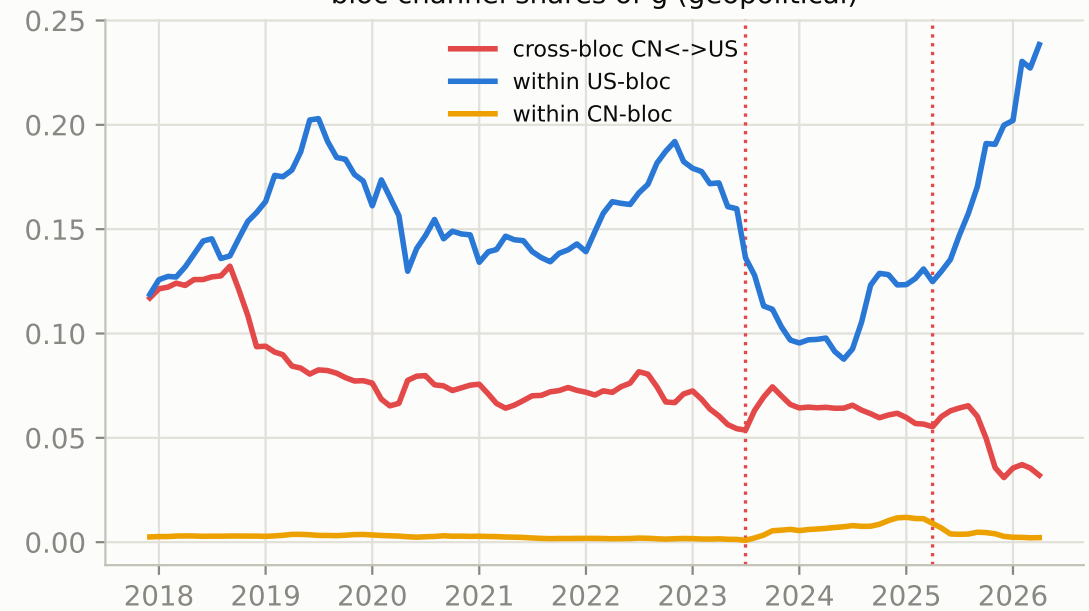
Left: hub-overlap fragmentation (range-compressed under the anchored basis; mainly a diagnostic). Right: bloc channel shares -- the cross-bloc CN-US share falls ~70% over the sample in two legs (2019, then 2023 onward) with a plateau between.

AI compute (3 codes) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)



# Joint concentration decomposition -- AI compute

3. Concentration & fragmentation

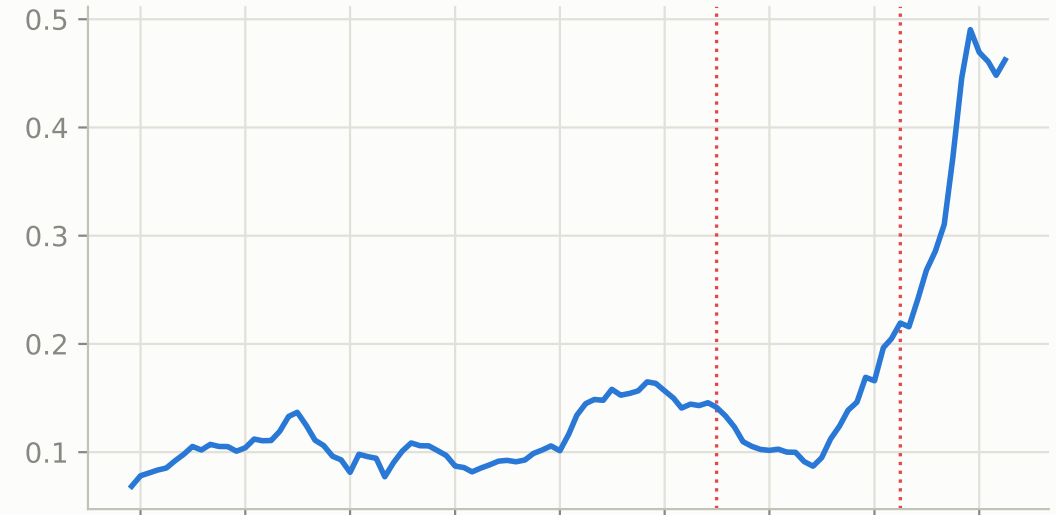
$\text{HHI\_flat} = \text{HHI\_F} \times (\mu_a \mu_b + \text{alignment covariance})$ , exact identity. Alignment covariance is mildly positive from 2017, peaks ~2019, decays through the AI era: no compounding-concentration signature.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

linkage concentration HHI\_F



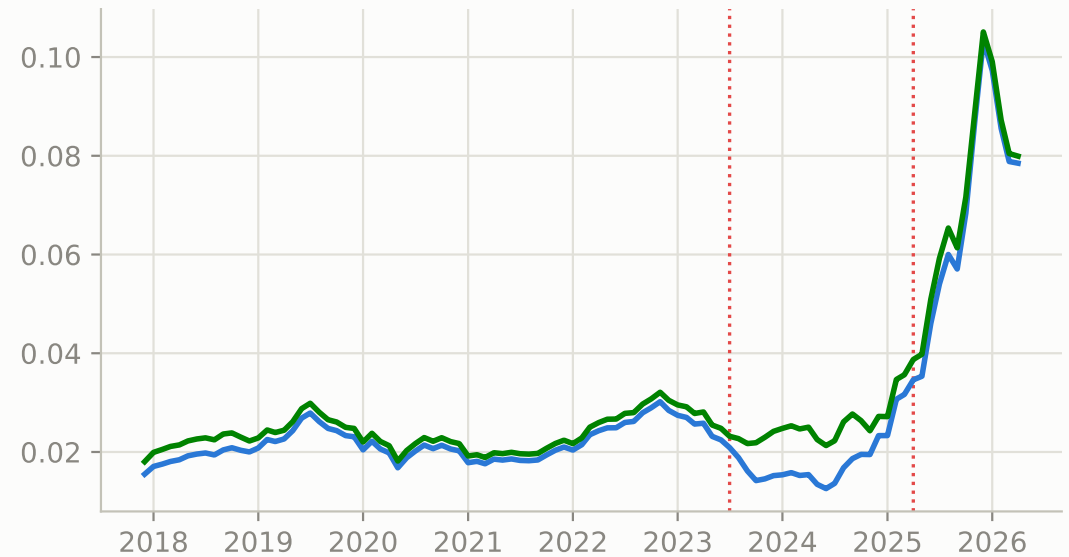
linkage-weighted means  $\mu_a * \mu_b$



alignment covariance  $\text{Cov}(a,b)$



joint: HHI\_flat (blue) vs bilateral (green)



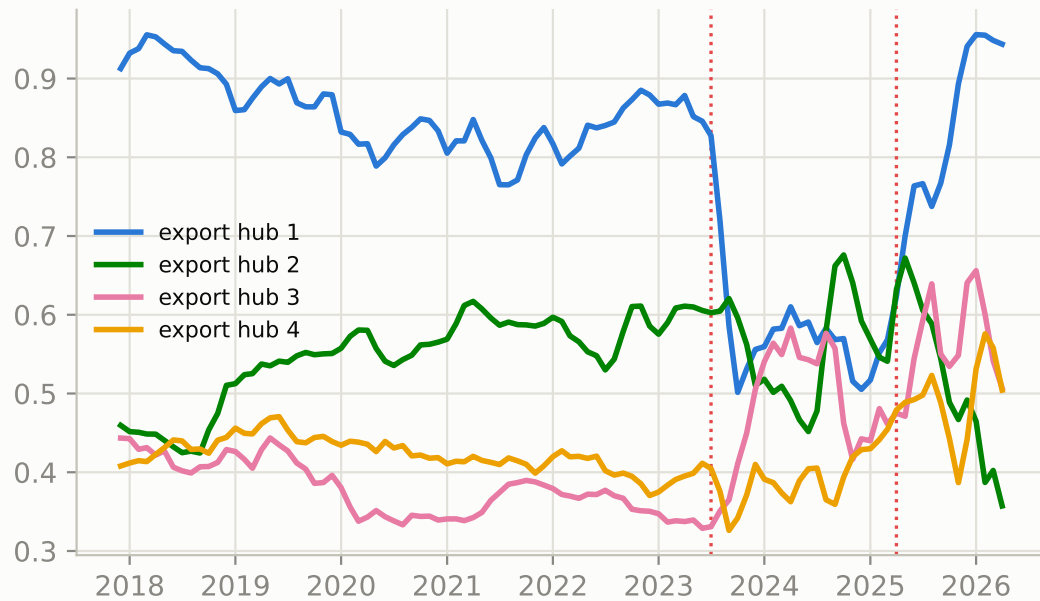
# Per-hub linkage concentration and basis diagnostic -- AI compute

3. Concentration & fragmentation

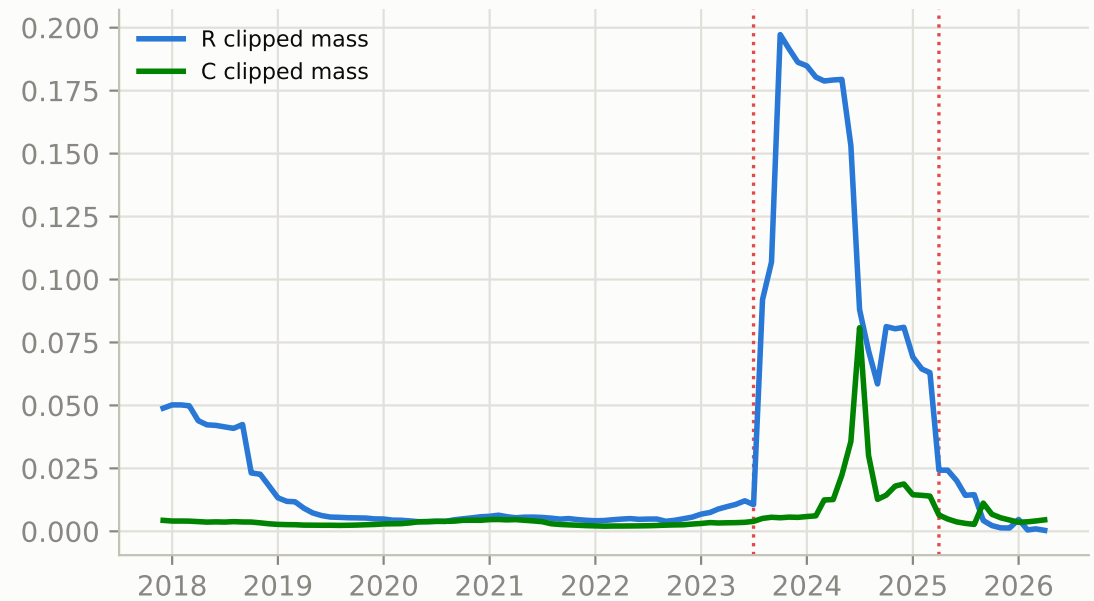
Left: does an export hub sell into one import program or several. Right: clipped negative mass (basis quality), elevated during the 2023-24 regime transition.

## AI compute (3 codes) — linkage structure and basis diagnostics

per-hub linkage HHI (1 = sells to a single import hub)



negative mass clipped (basis-quality diagnostic)

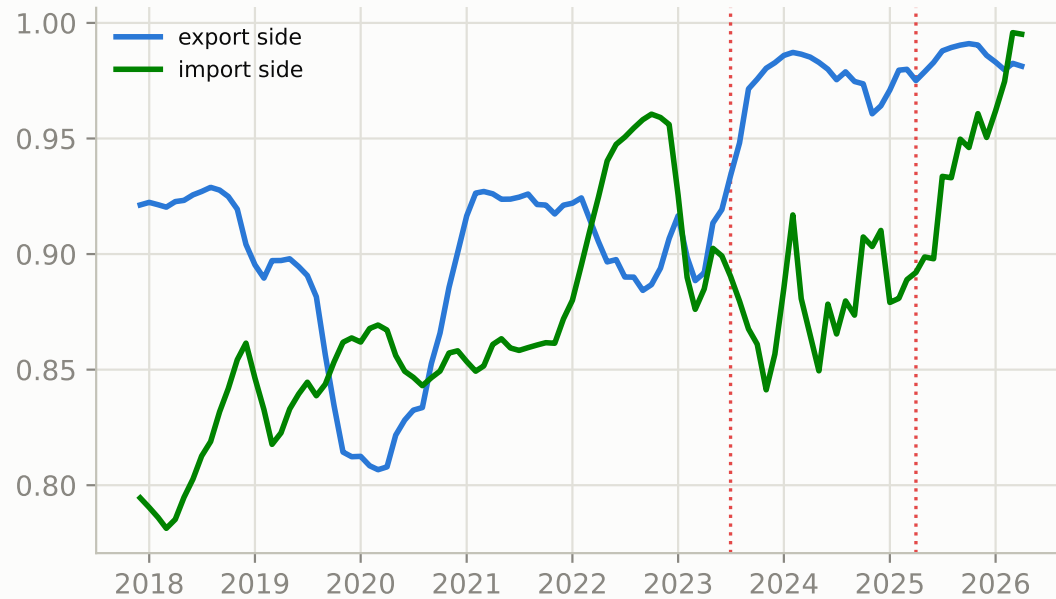


# Fragmentation -- baseboards (847180)

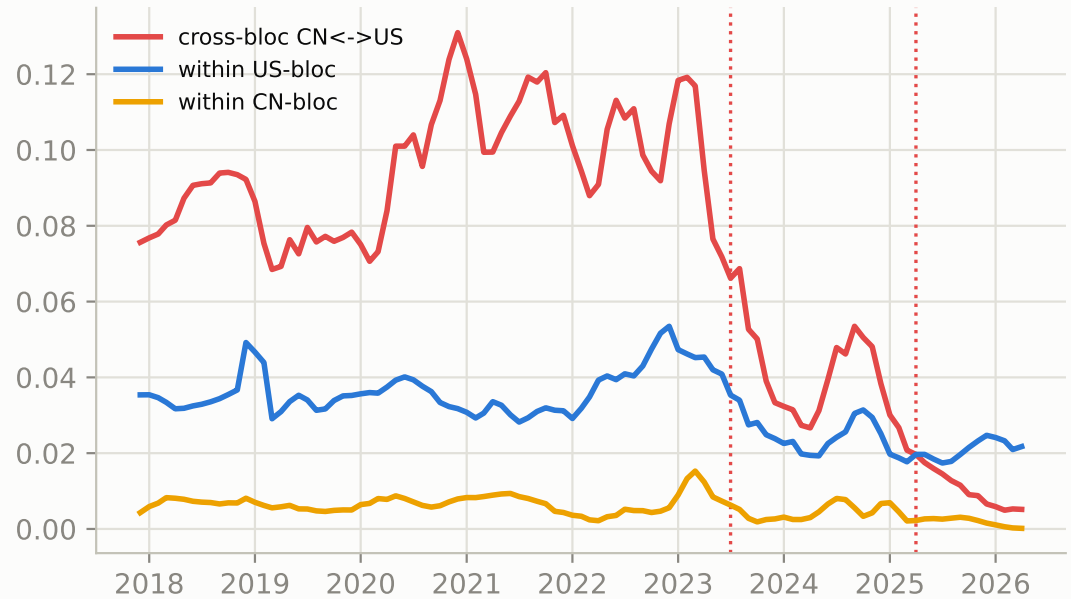
The sharpest decoupling of the three codes: the cross-bloc share collapses ~20x from its 2021 peak while linkage concentration HHI\_F rises 0.09 -> 0.48.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)

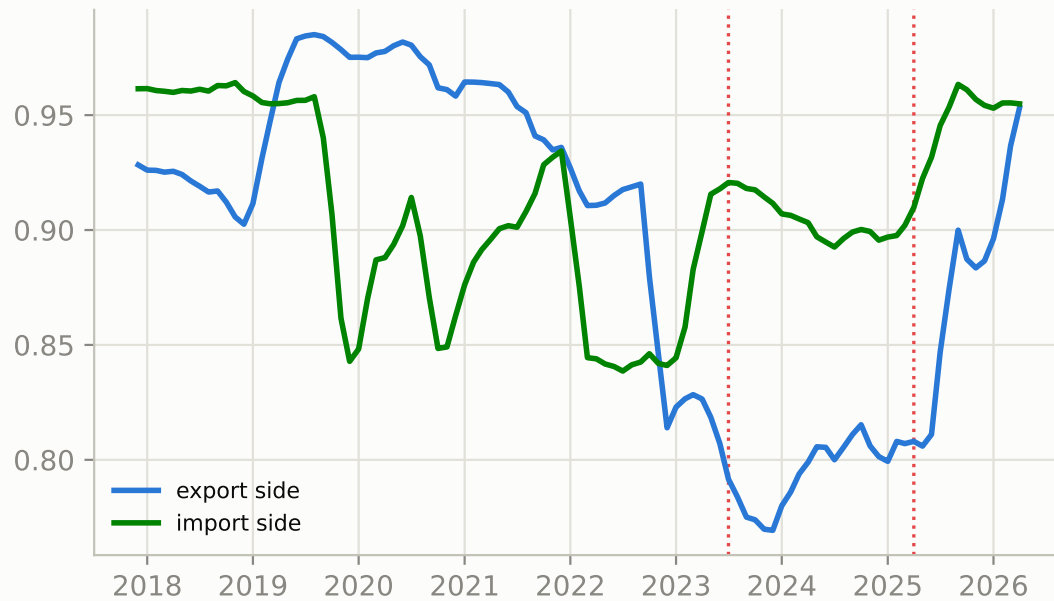


# Fragmentation -- parts (847330)

Cross-bloc share halves but remains the highest of the three codes: the parts layer stays the shared substrate.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)

